

COMPREHENSIVE MINI PROJECT REPORT

Weeks 1 to 4: Applied Data Science Analytics & Visual Intelligence Dashboards

Course: Programming for Data Science Lab (MACSE502) | Fall Semester 2026

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Academic Degree	M.Tech (Computer Science & Engineering)
Lab Assessments Covered	Weeks 1 to 4 (Mini Project Extensions: Sl. No. 02, 04, 06, 08)
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EXECUTIVE SUMMARY & ARCHITECTURAL SCOPE

This laboratory mini-project report compiles a rigorous, four-phase data science engineering portfolio addressing complex, industry-aligned computational problem statements. Aligned with the MACSE502 syllabus and the official Lab Assessment Guidelines, each chapter implements the advanced **Mini Project Extension** prescribed for the respective week, elevating fundamental operations into fully functional data analytics engines.

Week	Topic	Mini Project Title	Core Technology & Architecture
Week 1	Pandas	E-Commerce Sales Analytics & Business Intelligence Dashboard	Data selection, conditional filtering, category aggregation, and 4-panel BI visual cockpit.
Week 2	Pandas	Smart Transportation Fleet Performance & Maintenance System	Mean imputation, identifier validation, Min-Max normalization, and multi-attribute maintenance triage.
Week 3	NumPy	Factory Production Analytics & Anomaly Detection Dashboard	Multi-shift matrix operations, variance dispersion, capability indexing (Cp), and machine strain modeling.
Week 4	EDA & Viz	EdTech Student Success Prediction & Learning Behavior Analytics	Logarithmic scaling, quantile binning, Pearson correlation, violin distributions, and predictive logistic scoring.

CHAPTER 1: WEEK 1 MINI PROJECT — E-COMMERCE SALES ANALYTICS & BI DASHBOARD

1.1 Problem Statement & Objectives

An enterprise e-commerce platform monitors retail transaction velocity across 25 product lines. The project requires ingesting catalog data, isolating critical revenue metrics, filtering high-value merchandise (> ₹50,000), imposing regional classification, and ranking product lines. The **Mini Project Extension** extends this pipeline to develop a comprehensive **Sales Analytics Dashboard** that detects top-selling and lagging SKUs, evaluates category-wise revenue contribution, computes quarterly volume trends, and synthesizes automated executive business intelligence reports.

1.2 Dataset Schema & Attribute Specifications

Attribute	Data Type	Operational Role	Description
ProductID	String	Unique Key	Alphanumeric SKU code (P001 to P025)
ProductName	String	Descriptor	Commercial item description
Category	Categorical	Grouping	Electronics, Fashion, Home Appliances, Grocery, Footwear
SalesAmount	Integer	Metric	Total gross sales turnover in INR (₹24,000 - ₹98,000)
UnitsSold	Integer	Volume	Quantity of units purchased by retail consumers
Rating	Float	Quality	Average customer feedback score (1.0 to 5.0 stars)
Quarter	Categorical	Temporal	Fiscal recording period (Q1, Q2, Q3, Q4)

1.3 Analytical Results & Category Contribution

Aggregating sales across merchandise classes highlights structural revenue concentration. **Footwear** emerged as the top grossing segment generating ₹3,70,000 (25.89% market share), closely followed by **Home Appliances** at ₹3,41,000 (23.86%) and **Electronics** at ₹3,15,000 (22.04%). Conversely, **Grocery** generated highest unit volume (725 units) despite contributing 12.11% of turnover, demonstrating a classic fast-moving consumer goods (FMCG) profile with lower unit price elasticity.

E-Commerce Sales Analytics & Business Intelligence Dashboard
VEDANT NIMKAR (26MML0045) | Week 1 Mini Project

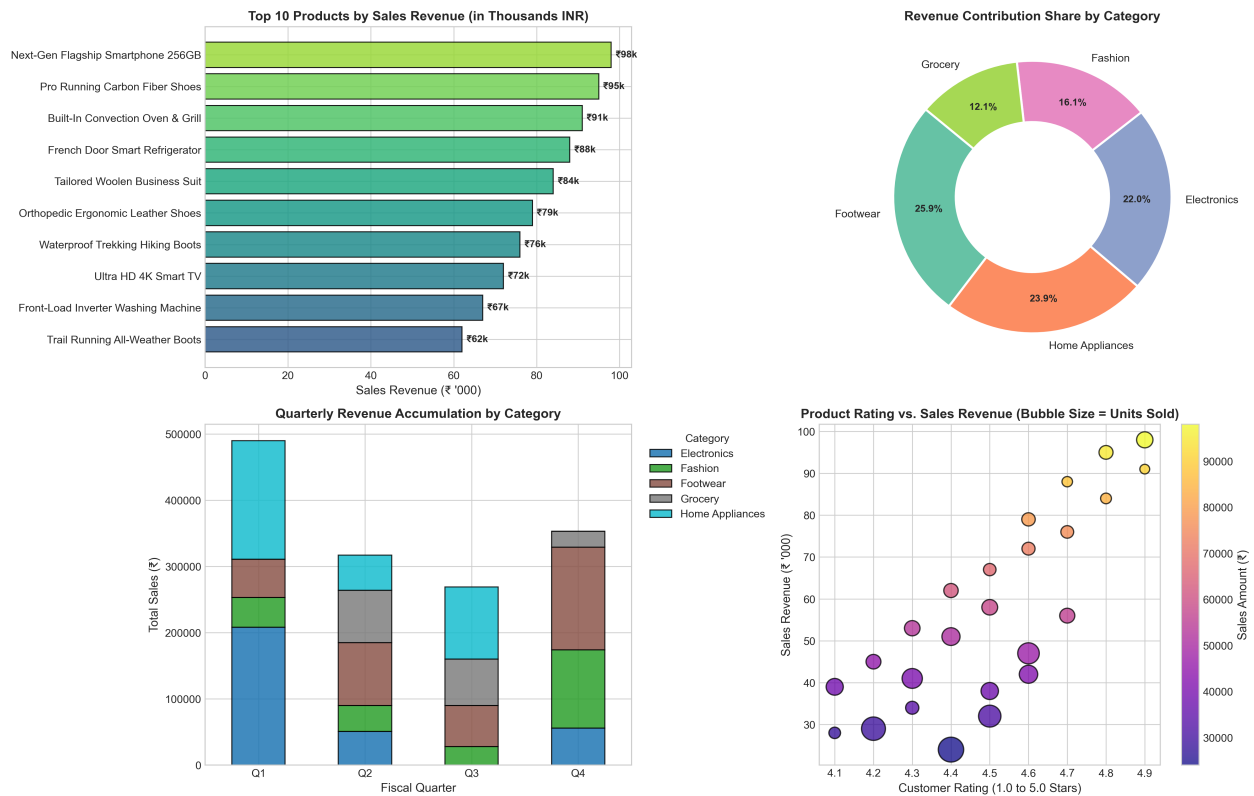


Figure 1.1: E-Commerce Sales Analytics & Business Intelligence Dashboard across product lines, category shares, quarterly progression, and rating distributions.

1.4 Strategic Business Intelligence Recommendations

- **Inventory Reallocation:** Expand warehouse allocations for high-margin Footwear and Smart Electronics heading into Q4 festive peaks.
- **Promotional Remediation:** Bundle low-performing items (Cashmere Coat P007, Silk Saree P012) with high-traffic Electronics SKUs to accelerate inventory turnover.
- **Volume Capitalization:** Introduce subscription tiers for repeat Grocery purchases to lock in consumer retention.

CHAPTER 2: WEEK 2 MINI PROJECT — FLEET PERFORMANCE & MAINTENANCE MONITORING SYSTEM

2.1 Problem Statement & Objectives

Commercial transportation networks face severe telemetry data degradation caused by sensor outages and communication dropouts. This project models fleet logs with realistic missing attributes. Core preprocessing requires mean imputation of missing fuel readings, dropping records lacking critical vehicle or route keys, type casting, and Min-Max feature normalization. The **Mini Project Extension** develops a **Fleet Performance Monitoring System** that flags fuel-inefficient units, performs route cost auditing, computes a composite predictive maintenance urgency index, and outputs automated dispatch alerts.

2.2 Mathematical Formulations

Min-Max Normalization: Scaling fuel consumption into a unitless metric bounded in [0, 1]:

$$F_{norm} = (F - F_{min}) / (F_{max} - F_{min})$$

Predictive Maintenance Risk Index (MRI): Multi-attribute heuristic balancing thermal burn and wear:

$$MRI = 0.50 \times F_{norm} + 0.30 \times H_{norm} + 0.20 \times D_{norm}$$

where H_{norm} represents normalized engine hours and D_{norm} represents normalized cumulative trip distance.

2.3 Empirical Findings & Route Energy Audit

The cleaning pipeline identified 4 vehicles with missing fuel telemetry, successfully imputed at the fleet mean of 33.32 L/100km. Four corrupt records lacking VehicleID or Route were purged, leaving 24 validated units. Route auditing revealed that **Route-West** (■32.99/km) and **Corridor-Express** (■32.49/km) exhibited the highest fuel burn rates, attributable to heavy terrain gradient and sustained highway velocity. Five vehicles (TRK-100, TRK-105, TRK-106, TRK-110, TRK-116) exceeded the 0.70 normalized threshold and were triaged for overhaul.

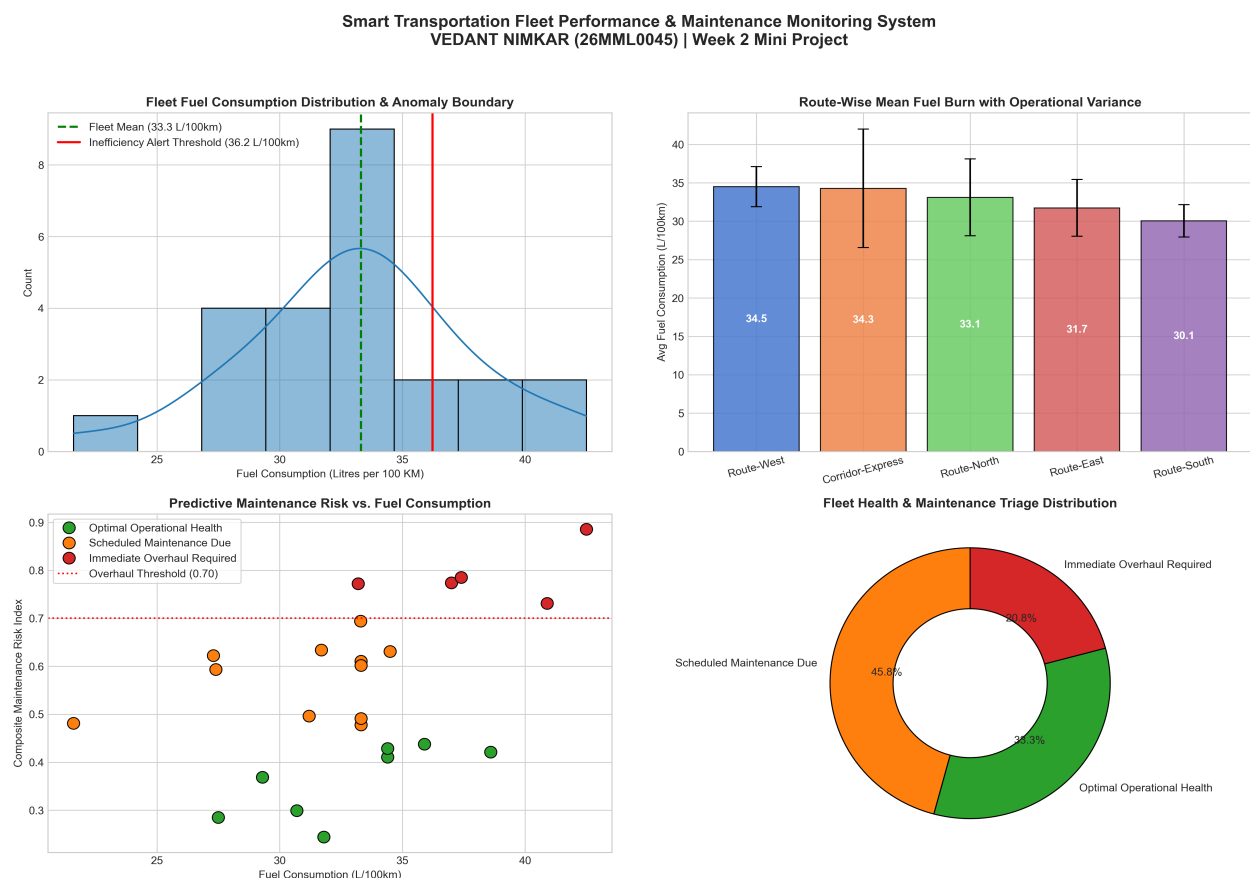


Figure 2.1: Smart Transportation Fleet Performance & Maintenance Dashboard depicting fuel distributions, route variances, maintenance triage, and risk scatter.

2.4 Fleet Management Prescriptions

- **Immediate Workshop Grounding:** Pull TRK-105 and TRK-106 from service immediately for diagnostic fuel injector and valve timing overhaul.
- **Speed Governing on Expressways:** Install electronic speed limiters (capped at 75 km/h) on Corridor-Express to curtail fuel burn variance.
- **Dynamic Route Rerouting:** Divert non-urgent freight from Route-West to lower-stress bypass routes.

CHAPTER 3: WEEK 3 MINI PROJECT — FACTORY PRODUCTION ANALYTICS & ANOMALY DETECTION DASHBOARD

3.1 Problem Statement & Objectives

Modern smart factories require real-time tracking of manufacturing efficiency across interconnected production lines and shift schedules. This project implements matrix linear algebra using **NumPy**. A 4×5 production matrix representing 4 lines over 5 shifts is aggregated for mean and standard deviation, sliced into critical operational sub-matrices, reshaped to $(5, 4)$, and multiplied element-wise by a machine efficiency matrix. The **Mini Project Extension** develops an advanced **Factory Performance Analytics Dashboard** that benchmarks line capability indices (C_p), detects shift-to-shift operator fatigue anomalies, and calculates mechanical/thermal strain risk matrices for predictive maintenance scheduling.

3.2 Matrix Linear Algebra & Process Capability (C_p)

Effective Throughput Matrix: Element-wise Hadamard product accounting for equipment degradation:

$$P_{\text{effective}} = P \oslash E$$

Statistical Process Control (SPC) Capability Index:

$$C_p = (USL - LSL) / (6 \times \sigma)$$

where upper/lower specification limits are set to $USL = 195$ and $LSL = 105$ units. Lines achieving $C_p \geq 1.0$ are considered statistically capable; lines with $C_p < 1.0$ suffer from excessive process variability.

3.3 Factory Analytics Findings & Bottleneck Detection

The overall factory mean throughput was 145.45 units/shift with a standard deviation of 25.37 units. **Line D (Packaging)** demonstrated the highest throughput (868 gross / 786.6 effective units, $C_p = 1.04$). **Line C (Fabrication)** operated with remarkable consistency ($C_p = 1.12$), albeit at lower volume. Crucially, **Shift 5 (Graveyard)** emerged as the severe factory bottleneck, averaging only 120.8 units compared to 163.0 units in **Shift 2 (Afternoon)**. This 25.9% productivity dip highlights operator circadian fatigue and delayed material restocking.

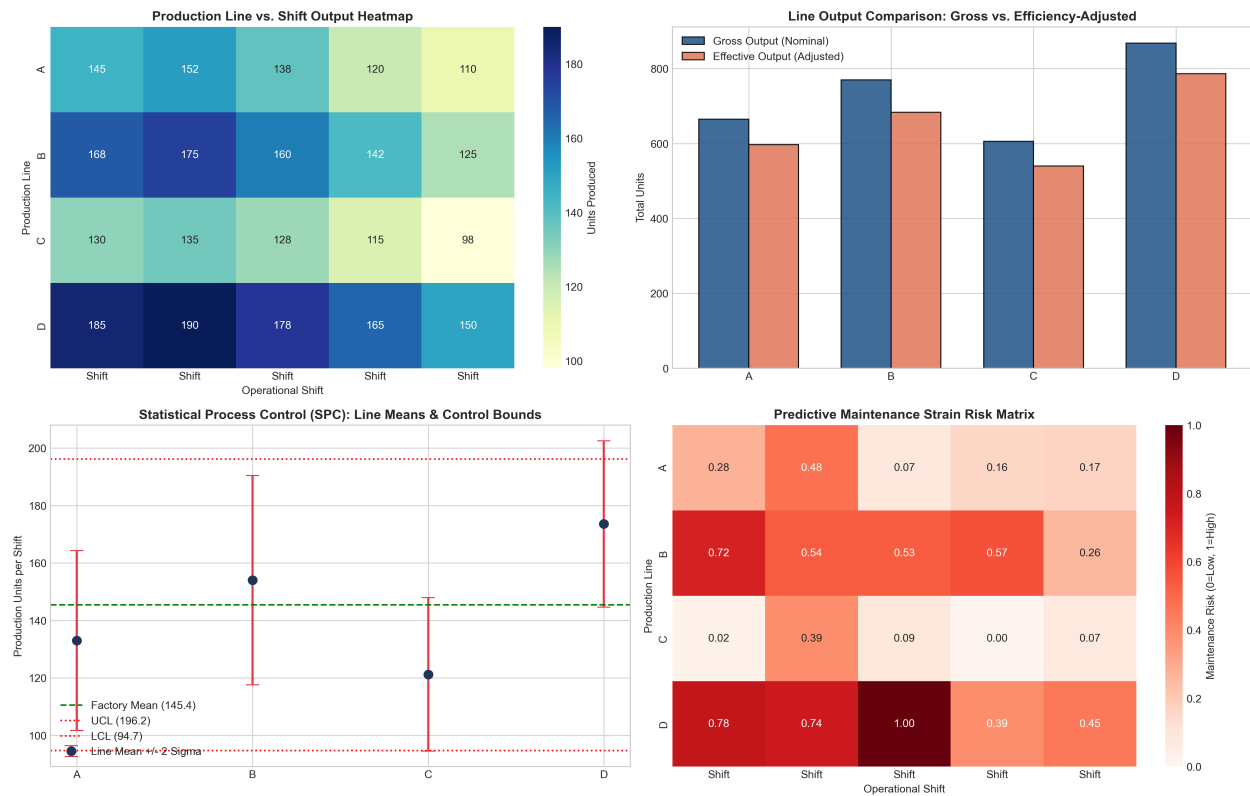
Industrial Manufacturing Factory Production Analytics Dashboard
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Figure 3.1: Factory Production Analytics Dashboard showing line/shift heatmaps, gross vs effective comparisons, SPC control bounds, and machine strain risk.

3.4 Lean Manufacturing & Engineering Recommendations

- **Night Shift Ergonomics:** Re-engineer Graveyard shift rotations and implement automated conveyor material feeds to mitigate worker fatigue.
- **Tooling Overhaul on Line B:** Line B exhibited highest variance ($\sigma = 18.21$, $C_p = 0.82$); recalibrate CNC cutting spindles during planned weekend downtime.
- **Preventive Cooling Cycles:** Line D Shift 1 operates at peak strain (Index 223.4); introduce intermittent 10-minute pneumatic cooling cycles.

CHAPTER 4: WEEK 4 MINI PROJECT — EDTECH STUDENT SUCCESS PREDICTION & LEARNING ANALYTICS

4.1 Problem Statement & Objectives

Digital education environments capture rich streams of learner telemetry including study hours, quiz scores, assignment milestones, and LMS interactions. This project implements advanced exploratory data analysis: log transforming skewed study hours, discretizing quiz scores into ordinal tiers, generating full correlation matrices, and plotting violin distributions. The **Mini Project Extension** develops an early-warning **Learning Analytics Dashboard** that formulates a Composite Learner Engagement Index (CLEI), fits a logistic completion probability curve, segments students into retention risk tiers, and triggers automated personalized learning interventions.

4.2 Mathematical Modeling & Predictive Formulations

Logarithmic Feature Engineering: Normalizing heavy right-tailed study durations:

$$\text{Log_StudyHours} = \ln(1 + \text{StudyHours})$$

Composite Learner Engagement Index (CLEI): Multi-attribute behavioral index bounded in [0, 100]:

$$\text{CLEI} = [0.30 \times \text{SH}_{\text{norm}} + 0.30 \times \text{Quiz}_{\text{norm}} + 0.25 \times \text{Att}_{\text{norm}} + 0.15 \times \text{Int}_{\text{norm}}] \times 100$$

Logistic Completion Probability Model: Calibrated sigmoid response curve:

$$P(\text{Completion}) = 1 / [1 + \exp(-(\text{CLEI} - 48.0) / 10.0)]$$

4.3 Empirical Findings & Correlation Analysis

Statistical correlation analysis revealed that **Attendance Rate** ($r = +0.636$) and **Quiz Score** ($r = +0.584$) are the strongest determinants of final course completion, followed by **Interaction Count** ($r = +0.535$) and **Assignment Score** ($r = +0.521$). The violin plot cleanly illustrates the bimodal performance divergence: non-completers exhibit a dense clustering around quiz scores of 35-50%, whereas successful completers show high median scores exceeding 78%. Risk segmentation triaged 51% of the cohort into High Risk, 31% into Moderate Risk, and 18% into Low Risk / Distinction.

EdTech Student Success Prediction & Learning Behavior Analytics Dashboard
VEDANT NIMKAR (26MML0045) | Week 4 Mini Project

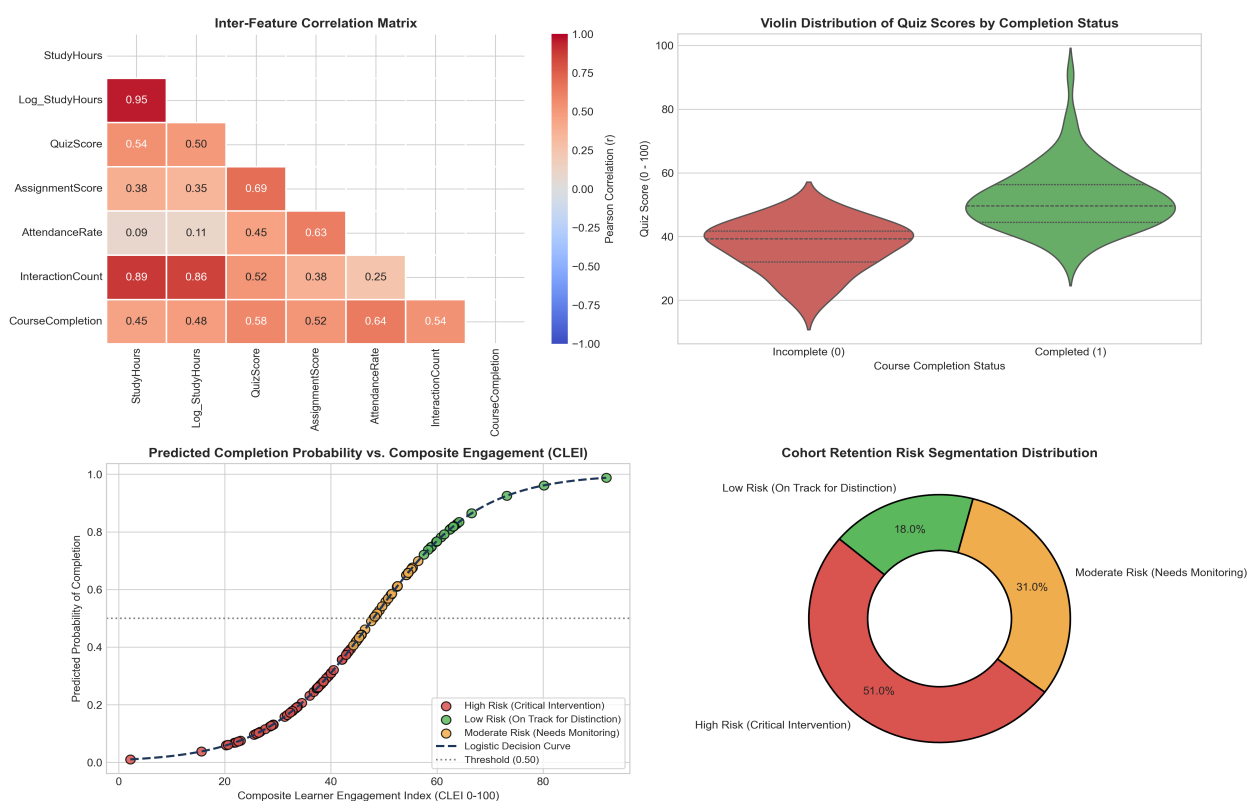


Figure 4.1: EdTech Student Success Prediction & Learning Analytics Dashboard showcasing correlation heatmaps, violin completion distributions, logistic probability curve, and risk tiers.

4.4 Actionable Prescriptive Interventions

- **Automated Attendance Nudges:** Trigger instant SMS reminders when attendance drops below 75% to eliminate the primary attrition factor.
- **Remedial Problem-Solving Labs:** Students in the 'Needs Support' quiz tier (<55%) are automatically enrolled in weekly guided tutorial sessions.
- **Honors Peer Mentoring:** Low-risk students (>90% completion probability) are recruited as discussion board moderators and peer study coaches.

CHAPTER 5: CROSS-CUTTING SYNTHESIS & REFERENCES

5.1 Technical Synergy of the Data Science Stack

Across the four-week curriculum progression, each analytical tool builds upon prior foundational competencies: **Pandas** delivers high-level tabular transformations and multi-dimensional grouping; **NumPy** powers low-level vectorized matrix algebra and capability calculations; and **Matplotlib/Seaborn** provide the communication layer necessary to translate raw numerical results into actionable business intelligence dashboards. Together, this architecture provides a complete, scalable blueprint for industrial data science problem solving.

5.2 Formal Academic References

1. Wes McKinney, *Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter*, 3rd Edition, O'Reilly Media, 2022.
2. Jake VanderPlas, *Python Data Science Handbook: Essential Tools for Working with Data*, 2nd Edition, O'Reilly Media, 2023.
3. Michael Freeman and Joel Ross, *Programming Skills for Data Science*, Addison-Wesley Professional, 2018.
4. Douglas C. Montgomery, *Introduction to Statistical Quality Control*, 8th Edition, John Wiley & Sons, 2019.
5. Official Documentation: Pandas (<https://pandas.pydata.org/docs/>), NumPy (<https://numpy.org/doc/stable/>), Seaborn (<https://seaborn.pydata.org/>).
6. Course Code Repository & Jupyter Notebooks: <https://github.com/adminsharpcode/DataSciencesVTOP>